

Redes convolucionales

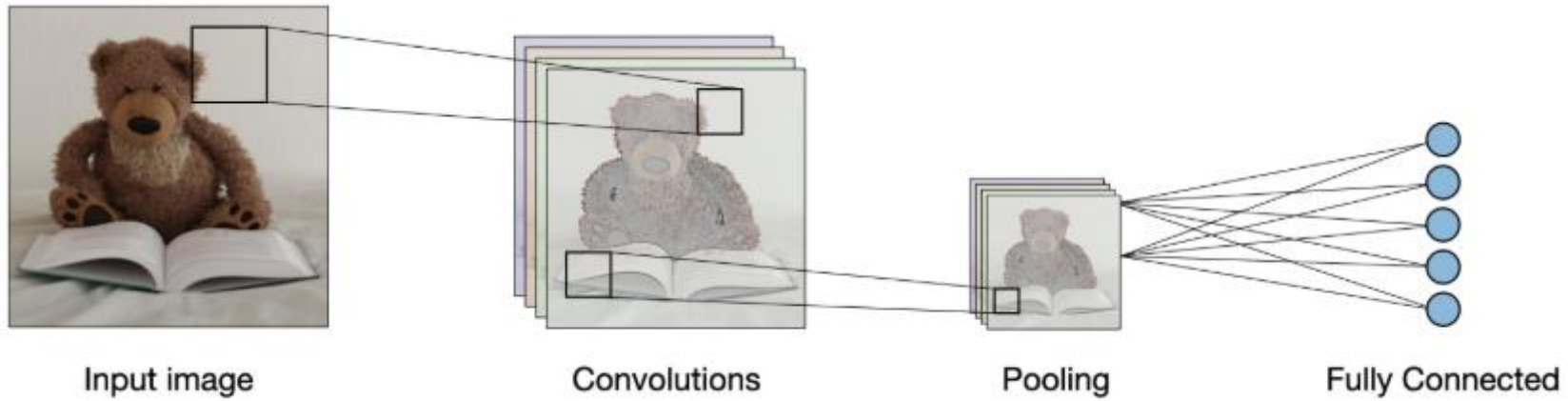
Juan Pajaro

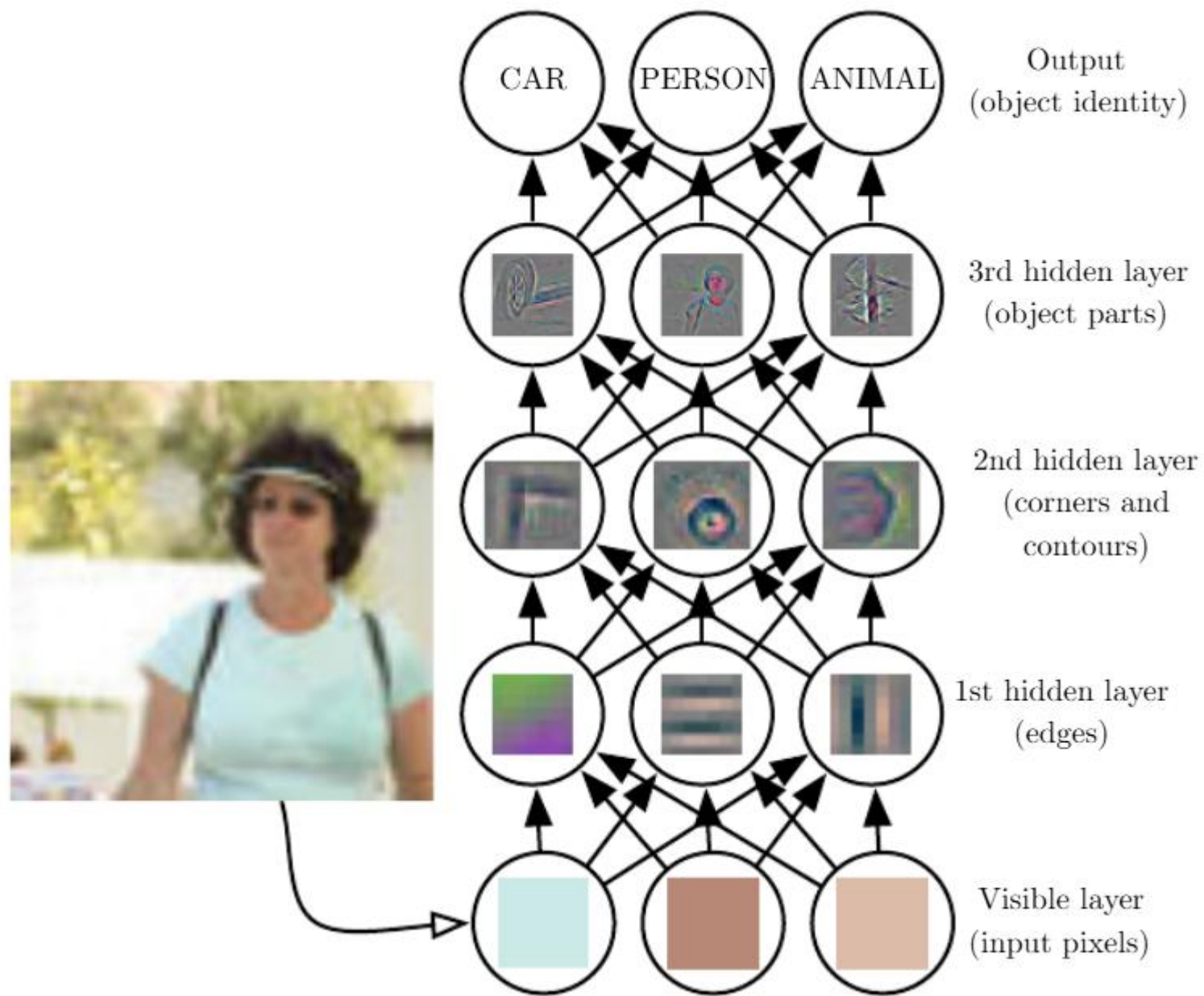
juanpajaro@javeriana.edu.co

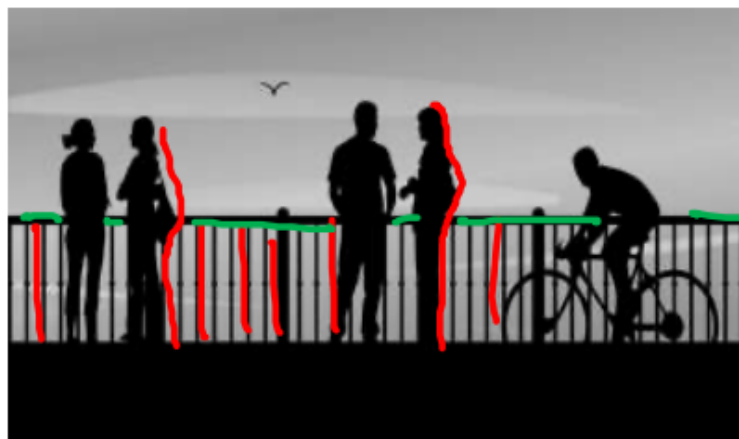
Agenda

- Redes convolucionales
- Gradientes integrados – técnicas de interpretación.

Redes convolucionales





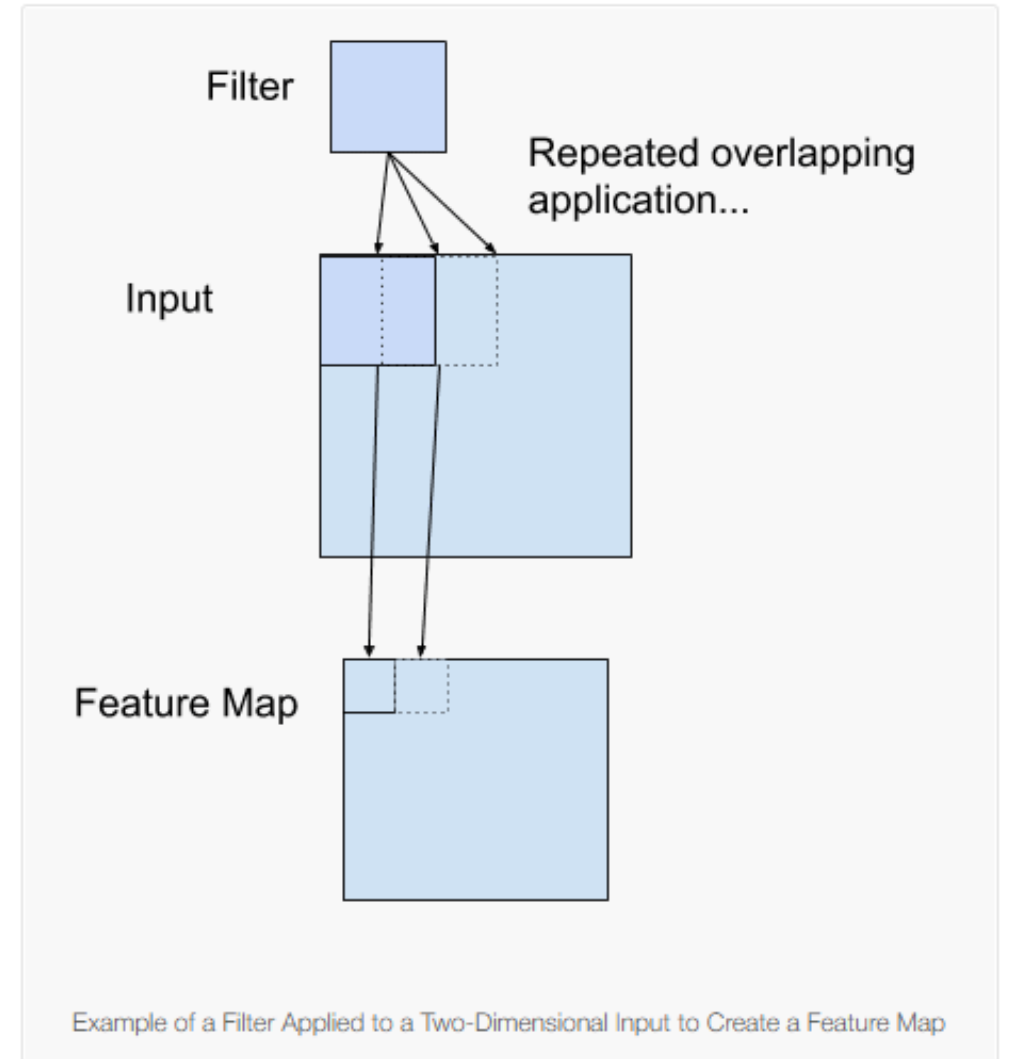
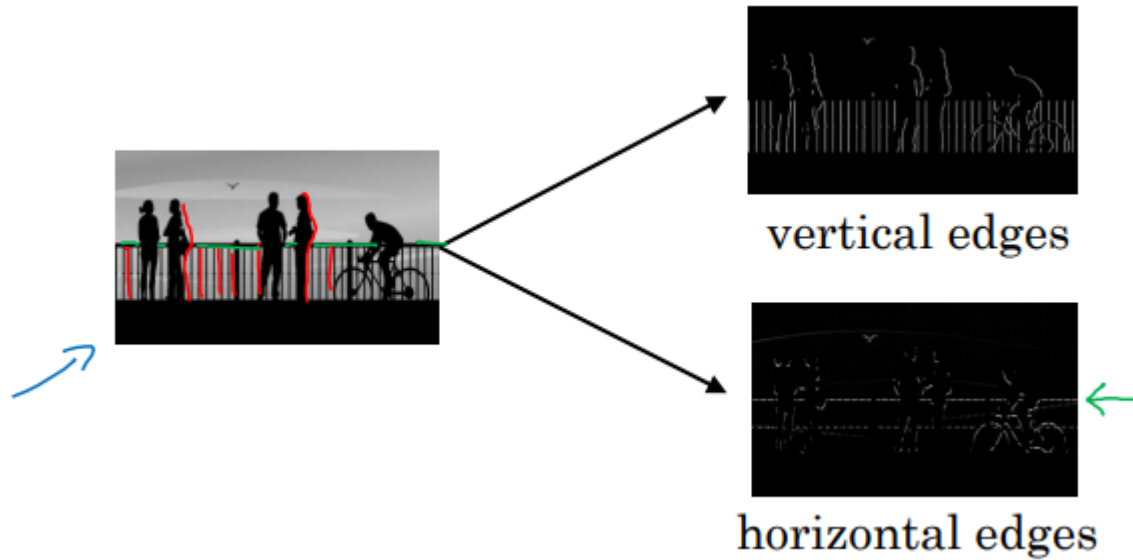


vertical edges

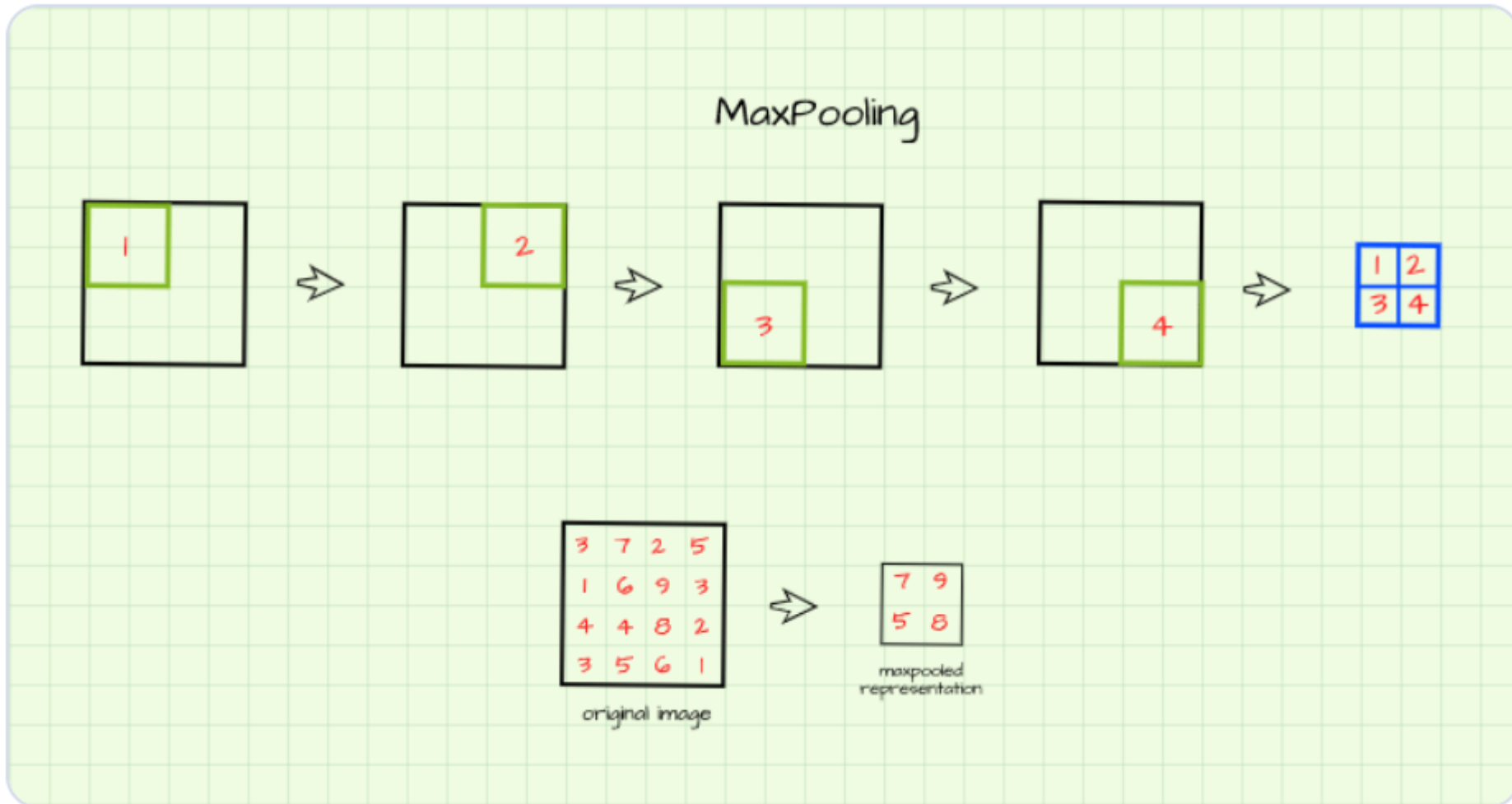


horizontal edges

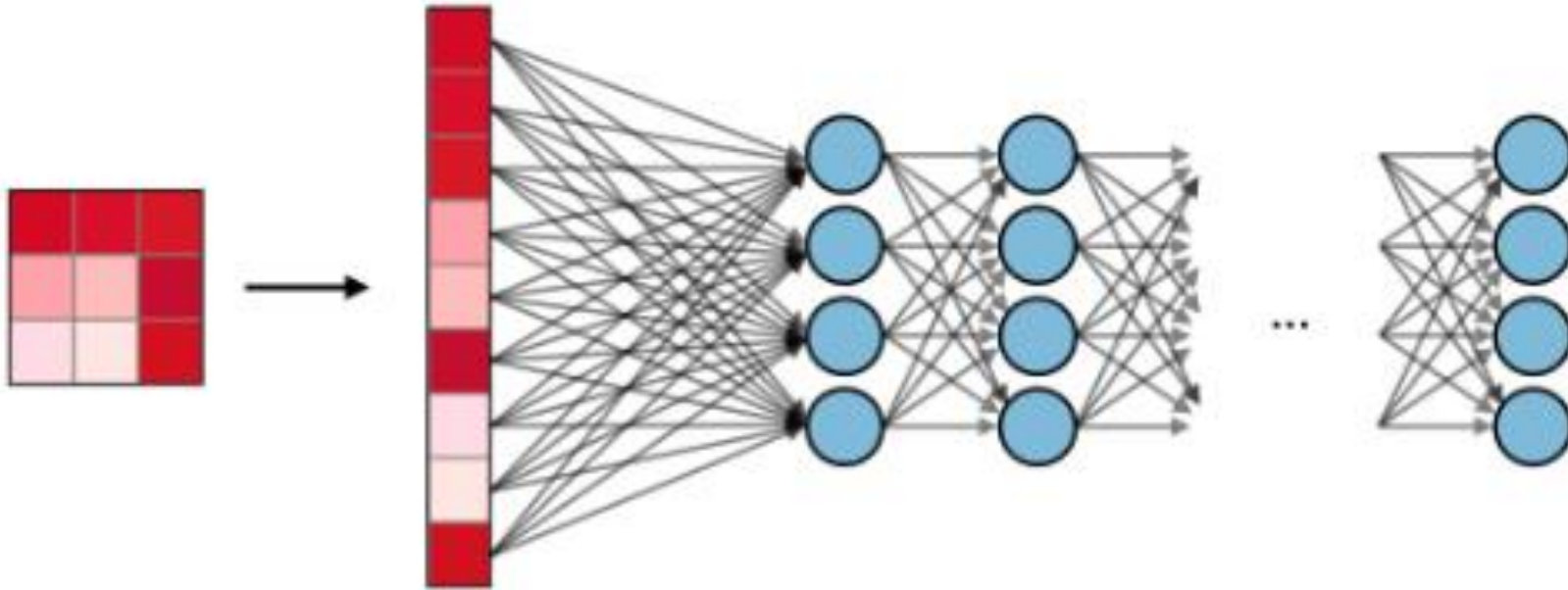
Convoluciones



Pooling

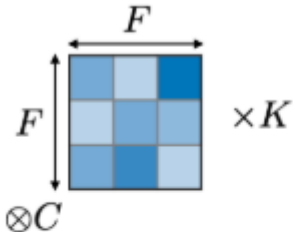
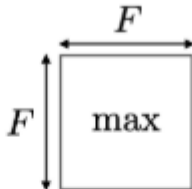
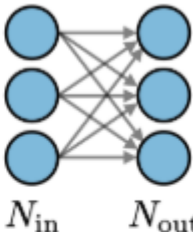


Fully connected



- Colab

Numero de parametros

	CONV	POOL	FC
Illustration			
Input size	$I \times I \times C$	$I \times I \times C$	N_{in}
Output size	$O \times O \times K$	$O \times O \times C$	N_{out}
Number of parameters	$(F \times F \times C + 1) \cdot K$	0	$(N_{in} + 1) \times N_{out}$
Remarks	• One bias parameter per filter • In most cases, $S < F$ • A common choice for K is $2C$	• Pooling operation done channel-wise • In most cases, $S = F$	• Input is flattened • One bias parameter per neuron • The number of FC neurons is free of structural constraints

- Gradientes integrados – técnicas de interpretación.
- <https://docs.seldon.ai/alibi-explain>

Referencias

- <https://stanford.edu/~shervine/teaching/cs-230/cheatsheet-convolutional-neural-networks/>